Questão 02

$$I = \int_0^1 \frac{1}{(1+x)^2}\,dx \Rightarrow f(x) = \frac{1}{(1+x)^2},\ 0 \leq x \leq 1$$

a) $X \sim U(0,1) \Rightarrow [0,1]$

$p\,U\,f(x) = 1,\ 0 \leq x \leq 1$

$$I = \int_0^1 \frac{1}{(1+x)^2}\,dx = E\left[\frac{1}{(1+x)^2}\right]$$

$$\hat{I}_{MC} = \frac{1}{N}\sum_{i=1}^{N} \frac{1}{(1+x_i)^2},\ x_i \sim U(0,1)$$

Primitiva:

$$\int (1+x)^{-2}\,dx = -(1+x)^{-x} + C$$

$$I = \left[-\frac{1}{1+x}\right]_0^1 = -\frac{1}{2} - (-1) = \frac{1}{2}$$

$$E[f(x)] = I = \frac{1}{2}$$

$$E[f(x)^2] = \int_0^1 \left(\frac{1}{(1+x)^2}\right)^2 dx = \int_0^1 \frac{1}{(1+x)^4} dx$$

Primitiva

$$\int (1+x)^{-4} dx = -\frac{1}{3}(1+x)^{-3} + C \therefore$$

$$E[f(x)^2] = \left[-\frac{1}{3(1+x)^3}\right]_0^1 \therefore -\frac{1}{3.2^3} - \left(-\frac{1}{3}\right)$$

$$E[f(x)^2] = \frac{7}{24}$$

$$var(f(x)) = E[f(x)^2] - E[f(x)]^2$$

$$Var(f(x)) = \frac{7}{24} - \left(\frac{1}{2}\right)^2 = \frac{1}{24}$$

$$Var(\hat{I}_{mc}) = \frac{1}{24N}$$

b) $I = \int_0^1 f(x)\,dx = \int_0^1 \frac{f(x)}{p(x)}\,p(x)\,dx = E_p\left[\frac{f(x)}{p(x)}\right]$

$p(x) \sim [0,1] \qquad f(x) > 0$

$f(x) = \frac{1}{(1+x)^2} > 0 \quad \therefore \; p(x) = c \cdot \frac{1}{(1+x)^2}$

$\int_0^1 p(x)\,dx = 1 \Rightarrow c\int_0^1 \frac{1}{(1+x)^2}\,dx = 1$

$c \cdot \frac{1}{2} = 1 \quad \therefore \; c = 2$

$p(x) = \frac{2}{(1+x)^2}, \quad 0 \leq x \leq 1$

$\int_0^1 \frac{2}{(1+x)^2}\,dx = 2 \cdot \frac{1}{2} = 1$

c) $I = \int_0^1 \frac{f(x)}{p(x)} p(x)\,dx = E_p\left[\frac{f(X)}{p(X)}\right]$

$\hat{I}_{IS} = \frac{1}{N}\sum_{i=1}^{N} \frac{f(Y_i)}{p(Y_i)}, \; Y_i \sim p$

$f(x) = \frac{1}{(1+x)^2}$ e $p(x) = \frac{2}{(1+x)^2}$

$\frac{f(x)}{p(x)} = \frac{1}{2}$ para todo $x \in [0,1]$

$\hat{I}_{IS} = \frac{1}{N}\sum_{i=1}^{N} \frac{1}{2} = \frac{1}{2}$, a integral é $I = \frac{1}{2}$

d) Estimador:

$\hat{I}_{MC} = \frac{1}{N}\sum_{i=1}^{N} \frac{1}{(1+X_i)^2}, \; X_i \sim U(0,1)$

Variância

$$Var(\hat{I}_{MC}) = \frac{1}{24N}$$

Existe um erro aleatório, que diminui conforme N aumenta

$\hat{I}_{is} = \frac{1}{2}$ e $Var(\hat{I}_{is}) = 0$

O método de Importance Sampling, com a escolha de $p(x)$, é bem melhor que Monte Carlo simples, pois

$$Var(\hat{I}_{is}) = 0 \text{ enquanto } Var(\hat{I}_{MC}) = \frac{1}{24N}$$

No Monte Carlo simples a estimativa oscila em torno de $\frac{1}{2}$

No Importance Sampling escolhido a estimativa sempre é $\frac{1}{2}$